1. Library Management System

Design a Java class to represent a library book.

Code:

```
import java.util.Scanner;
class LibraryBook {
    int bookId;
    String title, author;
    int copies = 20;
    void read() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Book ID: ");
        bookId = sc.nextInt();
        sc.nextLine();
        System.out.print("Title: ");
        title = sc.nextLine();
        System.out.print("Author: ");
        author = sc.nextLine();
    }
    void issueBook() {
        copies--;
    }
    void returnBook() {
        copies++;
    }
    void display() {
        System.out.println(bookId + " " + title + " " + author);
        System.out.println("Available Copies: " + copies);
    }

    public static void main(String args[]) {
```

```java
        LibraryBook b = new LibraryBook();
        b.read();
        b.issueBook();
        b.returnBook();
        b.display();
    }
}
```

Output:

Book ID: 101

Title: Java Programming

Author: James Gosling

101 Java Programming James Gosling

Available Copies: 20

2. Employee Salary Management

Design a Java class to represent an employee.

Code:

```java
import java.util.Scanner;
class Employee {
    int empId;
    String name, dept;
    double basic = 25000;
    void read() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Employee ID: ");
        empId = sc.nextInt();
        sc.nextLine();
        System.out.print("Name: ");
        name = sc.nextLine();
```

```
        System.out.print("Department: ");
        dept = sc.nextLine();
    }
    void grossSalary() {
        double hra = basic * 0.20;
        double da = basic * 0.10;
        double gross = basic + hra + da;
        System.out.println("HRA = " + hra);
        System.out.println("DA = " + da);
        System.out.println("Gross Salary = " + gross);
    }
    public static void main(String args[]) {
        Employee e = new Employee();
        e.read();
        e.grossSalary();
    }
}
```

Output:

Employee ID: 1001

Name: Sai

Department: IT

HRA = 5000.0

DA = 2500.0

Gross Salary = 32500.0

3. Student Fee Management System

Design a Java class to represent a student.

Code:

```
import java.util.Scanner;
class Student {
    int id;
    String name, course;
    double feeBalance = 50000;
    void read() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Student ID: ");
        id = sc.nextInt();
        sc.nextLine();
        System.out.print("Name: ");
        name = sc.nextLine();
        System.out.print("Course: ");
        course = sc.nextLine();
    }
    void payFee(double amount) {
        feeBalance -= amount;
    }
    void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Course: " + course);
        System.out.println("Remaining Fee: " + feeBalance);
    }
    public static void main(String args[]) {
        Student s = new Student();
        s.read();
```

```
        s.payFee(10000);
        s.display();
    }
}
```

Output:

Student ID: 101

Name: Mounika

Course: CSE

ID: 101

Name: Mounika

Course: CSE

Remaining Fee: 40000.0

4. Electricity Bill Management

Design a Java class to represent a consumer.

Code:

```
import java.util.Scanner;
class Consumer {
    int cno, units;
    String name, type;
    void read() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Consumer No: ");
        cno = sc.nextInt();
        sc.nextLine();
        System.out.print("Name: ");
        name = sc.nextLine();
        System.out.print("Type (Domestic/Commercial): ");
        type = sc.nextLine();
        System.out.print("Units: ");
```

```
        units = sc.nextInt();
    }
    void calculateBill() {
        double bill = 0;
        if(type.equalsIgnoreCase("Domestic"))
            bill = units * 5;
        else if(type.equalsIgnoreCase("Commercial"))
            bill = units * 8;
        else {
            System.out.println("Invalid Connection Type");
            return;
        }
        System.out.println("Bill Amount = " + bill);
    }
    public static void main(String args[]) {
        Consumer c = new Consumer();
        c.read();
        c.calculateBill();
    }
}
```

Output:

Consumer No: 1001

Name: Ravi

Type (Domestic/Commercial): Domestic

Units: 100

Bill Amount = 500.0

5. Mobile Recharge System

Design a Java class to represent a mobile subscriber.

Code:

```
import java.util.Scanner;
class MobileSubscriber {
    long mobile;
    String name, provider;
    double balance = 500;
    void read() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Mobile Number: ");
        mobile = sc.nextLong();
        sc.nextLine();
        System.out.print("Name: ");
        name = sc.nextLine();
        System.out.print("Provider: ");
        provider = sc.nextLine();
    }
    void recharge(double amount) {
        balance += amount;
    }
    void deduct(double amount) {
        balance -= amount;
    }
    void display() {
        System.out.println("Updated Balance = " + balance);
    }
    public static void main(String args[]) {
        MobileSubscriber m = new MobileSubscriber();
        m.read();
```

```
        m.recharge(200);
        m.deduct(50);
        m.display();
    }
}
```

Output:

Mobile Number: 9876543210

Name: Sai

Provider: Airtel

Updated Balance = 650.0